Student Name: Husanpreet Kaur **UID: 24BCY70138**
Branch: AIT-CSE (Cyber Security) **Section/Group: 24AIT_KRG2**
Semester: 4 **Date of Performance: 01/04/2026**
Subject Name: Database Management System **Subject Code: 24CSH-298**

WORKSHEET 8

AIM: To design and implement a stored procedure that accepts gender as an input parameter and returns the total number of employees based on the given gender, thereby demonstrating procedural programming and parameterized logic in PL/SQL.

S/W Requirement: Database Management System:

- Oracle Database Express Edition (Oracle XE)
- PostgreSQL Database

Database Administration Tool / Client Tool:

- Oracle SQL Developer (for Oracle XE)
- pgAdmin (for PostgreSQL)

OBJECTIVES:

To write and execute a PL/SQL stored procedure that dynamically accepts gender as an argument and computes the employee count corresponding to the given gender.

- Create tables and insert data

- Implement stored procedures

- Use IN, OUT, and INOUT parameters

PROBLEM STATEMENT:

Organizations often require flexible reporting mechanisms to analyze workforce distribution based on gender for compliance, diversity tracking, and internal analytics.

PRACTICAL/EXPERIMENT STEPS:

Design a stored procedure that:

1. Accepts gender as an input parameter

2. Counts the total number of employees matching the given gender
3. Returns the count as an output parameter

TABLE STRUCTURE:

```
create table employee_data(
emp_id serial primary key,
emp_name varchar(50),
gender varchar(10),
salary numeric(10,2)
);
```

Data Output | Messages | Notifications

```
CREATE TABLE

Query returned successfully in 168 msec.
```

```
insert into employee_data(emp_name,gender,salary) values
('Amit','Male',30000),
('Riya','Female',35000),
('John','Male',28000),
('Sneha','Female',40000)
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 | Page No: 1 of 1

	emp_id [PK] integer	emp_name character varying (50)	gender character varying (10)	salary numeric (10,2)
1	1	Amit	Male	30000.00
2	2	Riya	Female	35000.00
3	3	John	Male	28000.00
4	4	Sneha	Female	40000.00

1. PROBLEM STATEMENT – Procedure Implementation

Create a procedure to count employees based on gender.

PROGRAM:

```
CREATE OR REPLACE PROCEDURE count_emp(
    IN p_gen VARCHAR(20),
    OUT count_emp INT,
    INOUT status VARCHAR
)
AS
$$
BEGIN
    SELECT COUNT(*) INTO count_emp
    FROM employee_data
    WHERE gender = p_gen;

    status := 'SUCCESS';
END;
$$ LANGUAGE plpgsql;
```

OUTPUT:

```
-- calling
    DO
$$
DECLARE
    GEN VARCHAR(20) := 'Male';
    COUNT_BY_GEN INT;
    STATUS VARCHAR(20) := 'Fail';
BEGIN

    CALL COUNT_EMP(GEN, COUNT_BY_GEN, STATUS);

    RAISE NOTICE 'YOUR GENDER IS % YOUR COUNT IS % AND STATUS IS %',
        GEN, COUNT_BY_GEN, STATUS;

END;
$$;
```

Data Output | Messages | Notifications

```
NOTICE:  YOUR GENDER IS Male YOUR COUNT IS 2 AND STATUS IS SUCCESS
DO

Query returned successfully in 75 msec.
```

LEARNING OUTCOMES:

1. Understand the structure and components of a stored procedure
2. Implement **IN** and **OUT** parameters in PL/SQL procedures
3. Pass dynamic input values to stored procedures
4. Apply aggregate functions (like COUNT) within procedures

5. Develop reusable database logic for real-world business scenarios used in companies like CData and Toddle

CONCLUSION:

The experiment successfully demonstrated the design and implementation of a parameterized PL/SQL stored procedure to calculate employee count based on gender. By using IN and OUT parameters along with aggregate functions, dynamic and reusable database logic was achieved. This method improves efficiency in data retrieval and supports real-world organizational requirements such as workforce analysis and reporting.